Bishwajit Garai

Python Backend Developer | GenAI & LLM Ops Specialist | RAG Architect
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PROFESSIONAL SUMMARY

Python Backend Developer with **4.0+ years** of experience building scalable, **production-grade backend systems** and **GenAI-powered** applications. Specialized in **FastAPI, Flask, LLMOps, RAG architectures, autonomous agents, AI chatbots, and hybrid search** systems using **OpenSearch, Vespa, and Vector DBs**. Strong background in **microservices, async processing, Redis, and cloud-ready** deployments.

Technical Skills

Languages & Backend: Python, FastAPI, Django, Flask, REST, GraphQL, WebSockets.
GenAI & LLM: LLMs (Gemini, OpenAI-compatible), RAG, Prompt Engineering, Autonomous Agents, AI Chatbots, LLM Ops
Search & Data: OpenSearch, Vespa, Vector DB, Hybrid Search, MySQL, SQLite, Redis, Pandas, NumPy
Infra & DevOps: Docker, Nginx, AWS, Git, CI/CD, Async Workers
Others: Web Scraping, Automation, Unit Testing, React, JavaScript.

Experience

Endpoint It Services Pvt Ltd | Software Developer Mumbai | Oct 2024 – Current

Leading **backend and GenAI developmen**t for multiple **AI products** using **FastAPI, Redis, MySQL, and LLMs**, including **MeetMemo, Novo Chat, and Archaea**. Built AI systems for real-time meeting capture, movie recommendations, code search, electronics suggestions, and podcast moment extraction. Implemented **scalable async pipelines and backend services for high-concurrency, low-latency processing**.

Adansa Solutions Pvt Ltd | Software Developer Kolkata | June 2022 - May 2024

Led the development of **Flask-based web applications**, implementing robust backend systems. Utilized **Pandas, Numpy** for efficient **ETL** processes, ensuring **smooth data transformation and integration**. Collaborated with cross-functional teams to **optimize system performance**, **improve data workflows**, and deliver solutions that enhanced overall project efficiency.

Projects

Endpoint It Services Pvt Ltd

Archaea (AI Code Search Engine) : *Python, FastAPI, OpenSearch,, Embeddings*

- Built a **semantic and hybrid code search engine** enabling natural-language queries over large codebases.
- Implemented **lexical search, AST-based indexing, AST skeleton matching, and vector search** using CodeBERT and Sentence Transformers .
- Applied LLM-based re-ranking to improve result relevance and precision

MeetMemo : *Python, FastAPI , Generative AI , MySql*

- **AI-powered meeting assistant** with autonomous browser agents for real-time capture, transcription, summarization, sentiment analysis, and actionable insights.
- Built scalable FastAPI services with async processing to support concurrent meetings.

Novo Chat – Movie Query & Recommendation System : *Python ,FastAPI , Mysql , Agno, RAG*

- GenAI-powered conversational system for **natural-language movie search, recommendations, and contextual Q&A** .
- Implemented **RAG pipelines with vector embeddings** for accurate, context-aware responses.

Others Projects :**Rettlo**(Electronics Recommendation System), **Podcast Magic**(AI Podcast Moment Capture)

Adansa Solutions Pvt Ltd

Simplified Excel Upload : *Python , Pandas , Flask , Calamine*

- Built REST endpoints for Excel upload, validation, and bulk data ingestion
- Optimized the **upload pipeline to ensure seamless processing** and **validation of massive** customer datasets. Implemented performance improvements to manage high volumes of data efficiently, **reducing processing time by 40%**

Other Projects: Payroll Management System , Fixed Asset Depreciation etc.

Education

Mallabhum Institute of Technology, Bankura, WB WB, India
Btech | Computer Science and Engineering June 2020 - July 2023
CGPA: 83.7

Raja Ranjit Kishore Govt Polytechnic, Jhargram, WB WB, India
Diploma of Higher Education | Computer Science June 2017 - June 2020
CGPA: 77.6